

PHANIDHAR AKULA

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EDUCATION

- Miami University:** *M.S. Computer Science, GPA: 3.86/4.00* Aug 2024 – Aug 2026
Relevant Coursework: Generative AI, Advanced Database Systems, Machine Learning, Software Testing
Publications: 2 first-author manuscripts under review at IEEE T-ITS and Frontiers (2026)
Leadership: President, Graduate Students of Color Association (110+ members) | Elected Graduate Student Representative, University Graduate Council | Program Lead, NAE Grand Challenges Scholars Program
- Malla Reddy University:** *B.Tech Computer Science* Aug 2020 – May 2024
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems

SKILLS

- TypeScript, JavaScript, React, Redux, Next.js, Tailwind CSS, Node.js, REST APIs, Capacitor (iOS/Android)
- Python, Java, C++, FastAPI, PostgreSQL, MySQL, Supabase, OAuth, LLMs, RAG, AWS, Docker, CI/CD

EXPERIENCE

- Computer Science Researcher:** *Miami University* Aug 2024 – Jul 2026
- Developed SimForge, an open-source Python framework that benchmarks 3 traffic simulators under identical conditions from one input, with Docker, CI, and 668 pytest tests so every run reproduces exactly.
 - Optimized trip routing, one travel path per simulated resident, from 142 hours to 7, a 20x speedup, by caching each route once in Python and parallelizing across 16 cores of a supercomputer node.
 - Debugged Cityscape, an open-source C++ digital twin city simulator, tracing a data-loading bug to a single filter condition; 3 merged PRs restored 493K buildings and cut invented-home placements from 70% to 6%.
 - Designed a 16-configuration Python study across 5 cities on a supercomputer cluster, tuning how far the router searches for routes and improving worst-case commute-time accuracy from 81% to 92%.
- Software Engineer Intern:** *Smart Pathfinders Overseas Education* Jan 2024 – May 2024
- Shipped a private client portal in React and Redux, giving 180+ enrolled students one place to track applications, upload documents, and reach counselors instead of emailing staff for updates.
 - Implemented Python/MySQL REST APIs and data models behind role-based access control across 3 user roles, so each student saw only their own records while staff managed the full caseload from one dashboard.
- Full-Stack Developer:** *Freelance – E-Commerce & Fitness Startup Clients* May 2023 – Dec 2023
- Delivered production web applications, building React and Redux interfaces against Python/MySQL REST APIs and deploying to AWS, replacing manual spreadsheet workflows for 300+ daily users.
 - Migrated 10K+ records from client Excel workbooks into MySQL with a Python ETL script, validating row counts and key fields against the source so both clients cut over in one pass with no manual re-entry.

PROJECTS

- LumiAI:** *AI Tutoring Platform – studywithlumi.com* Oct 2025 – Present
- Built and shipped a production React/TypeScript platform spanning document-grounded chat, quizzes, flashcards, spaced repetition, and voice tutoring, serving 220+ students with 3K+ AI interactions.
 - Engineered a serverless LLM inference layer on Vercel with token-verified requests, response streaming, and prompt caching, blocking 100% of anonymous inference calls and serving 87% of input tokens from cache.
 - Architected the full-stack data layer on PostgreSQL with database-enforced row-level security across all 13 user tables, Google OAuth, and signed storage, so no query path can return another user's data.
 - Extended the platform to iOS and Android using Capacitor, reusing 95% of the existing TypeScript codebase in native shells with platform-specific auth and storage and submitting both builds for store review.